

Dear Arm Recruitment Team,

I am writing to apply for the Graduate CPU Physical Implementation Engineer position in Sophia Antipolis. I am currently completing my B.Sc. in Electrical and Electronic Engineering at BRAC University, with graduation expected in September 2026. My academic and research work has increasingly focused on VLSI design, digital hardware, timing analysis, processor architecture, and performance optimization, making Arm's CPU physical implementation team a particularly exciting opportunity for me.

A major part of my recent research has involved 45 nm CMOS design and timing-sensitive hardware systems. I co-authored the IEEE ISVLSI 2026 paper, "*Variation-Aware SRAM Compute-in-Memory Boolean Architecture for Binary Neural Networks*," where I worked on an SRAM-based compute-in-memory architecture and studied how process variation and clock timing affect hardware reliability and system-level performance. I performed statistical timing analysis and investigated the relationship between clock margin, timing violations, energy consumption, and neural-network accuracy. This experience gave me a strong appreciation for the trade-offs between performance, power, timing margin, and hardware robustness.

I have also designed a 45 nm CMOS PLL, including a phase-frequency detector, charge pump, loop filter, VCO, and frequency divider, targeting approximately 4.8 GHz operation. Alongside my transistor-level work, I have completed digital-design projects involving RISC-V processor architecture, arithmetic accelerators, SRAM and memory systems, finite-state machines, and RTL-based hardware design. These projects have strengthened my understanding of critical paths, clocking, digital architecture, simulation, and the relationship between an RTL design and its eventual hardware implementation.

What particularly attracts me to this role is Arm's emphasis on Power, Performance, and Area optimization and its ownership of the implementation flow from synthesis through GDSII. I am especially interested in developing deeper expertise in synthesis, floorplanning, clock-tree implementation, place-and-route, static timing analysis, IR-drop analysis, and physical-design closure. The opportunity to work alongside CPU microarchitecture and standard-cell teams while identifying timing bottlenecks and evaluating architectural trade-offs closely matches the direction in which I want to develop my career.

Thank you for considering my application. I would welcome the opportunity to discuss how my background and interests align with Arm's CPU Physical Implementation team.

Sincerely,

Md. Tanvir Ahmed Tonmoy  
B.Sc. in Electrical and Electronic Engineering  
BRAC University  
Expected Graduation: September 2026